

Boston House Price Predicton Model -
By Satyam Sharma(055)

Code ▾

Hide

```
#Load libraries
install.packages(c("ggplot2", "corrplot", "randomForest"))
```

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

<https://cran.rstudio.com/bin/windows/Rtools/>

Hide

```
library(ggplot2)
library(corrplot)
library(randomForest)
```

Hide

```
#Load dataset
data <- read.table("C:\\Users\\Dell\\Downloads\\housing.csv", header = FALSE)
#fix columns
colnames(data) <- c(
  "CRIM", "ZN", "INDUS", "CHAS", "NOX", "RM", "AGE",
  "DIS", "RAD", "TAX", "PTRATIO", "B", "LSTAT", "MEDV"
)
dim(data)
```

```
[1] 506  14
```

Hide

```
head(data)
```

	CRIM <dbl>	ZN <dbl>	INDUS <dbl>	CHAS <int>	NOX <dbl>	RM <dbl>	AGE <dbl>	DIS <dbl>	RAD <int>
1	0.00632	18	2.31	0	0.538	6.575	65.2	4.0900	1
2	0.02731	0	7.07	0	0.469	6.421	78.9	4.9671	2
3	0.02729	0	7.07	0	0.469	7.185	61.1	4.9671	2
4	0.03237	0	2.18	0	0.458	6.998	45.8	6.0622	3
5	0.06905	0	2.18	0	0.458	7.147	54.2	6.0622	3

	CRIM <dbl>	ZN <dbl>	INDUS <dbl>	CHAS <int>	NOX <dbl>	RM <dbl>	AGE <dbl>	DIS <dbl>	RAD <int>
6	0.02985	0	2.18	0	0.458	6.430	58.7	6.0622	3

6 rows | 1-10 of 14 columns

Hide

```
#Basic data check
str(data)
```

```
'data.frame':  506 obs. of  14 variables:
 $ CRIM      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
 $ ZN        : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
 $ INDUS     : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
 $ CHAS      : int   0 0 0 0 0 0 0 0 0 0 ...
 $ NOX       : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
 $ RM        : num  6.58 6.42 7.18 7 7.15 ...
 $ AGE       : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
 $ DIS       : num  4.09 4.97 4.97 6.06 6.06 ...
 $ RAD       : int   1 2 2 3 3 3 5 5 5 5 ...
 $ TAX       : num  296 242 242 222 222 222 311 311 311 311 ...
 $ PTRATIO   : num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
 $ B         : num  397 397 393 395 397 ...
 $ LSTAT     : num  4.98 9.14 4.03 2.94 5.33 ...
 $ MEDV      : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

Hide

```
summary(data)
```

CRIM		ZN		INDUS		CHAS		NOX	
Min.	: 0.00632	Min.	: 0.00	Min.	: 0.46	Min.	:0.00000	Min.	:0.3850
1st Qu.:	0.08205	1st Qu.:	0.00	1st Qu.:	5.19	1st Qu.:	0.00000	1st Qu.:	0.4490
Median	: 0.25651	Median	: 0.00	Median	: 9.69	Median	:0.00000	Median	:0.5380
Mean	: 3.61352	Mean	: 11.36	Mean	:11.14	Mean	:0.06917	Mean	:0.5547
3rd Qu.:	3.67708	3rd Qu.:	12.50	3rd Qu.:	18.10	3rd Qu.:	0.00000	3rd Qu.:	0.6240
Max.	:88.97620	Max.	:100.00	Max.	:27.74	Max.	:1.00000	Max.	:0.8710
RM		AGE		DIS		RAD		TAX	
Min.	:3.561	Min.	: 2.90	Min.	: 1.130	Min.	: 1.000	Min.	:187.0
1st Qu.:	5.886	1st Qu.:	45.02	1st Qu.:	2.100	1st Qu.:	4.000	1st Qu.:	279.0
Median	:6.208	Median	: 77.50	Median	: 3.207	Median	: 5.000	Median	:330.0
Mean	:6.285	Mean	: 68.57	Mean	: 3.795	Mean	: 9.549	Mean	:408.2
3rd Qu.:	6.623	3rd Qu.:	94.08	3rd Qu.:	5.188	3rd Qu.:	24.000	3rd Qu.:	666.0
Max.	:8.780	Max.	:100.00	Max.	:12.127	Max.	:24.000	Max.	:711.0
PTRATIO		B		LSTAT		MEDV			
Min.	:12.60	Min.	: 0.32	Min.	: 1.73	Min.	: 5.00		
1st Qu.:	17.40	1st Qu.:	375.38	1st Qu.:	6.95	1st Qu.:	17.02		
Median	:19.05	Median	:391.44	Median	:11.36	Median	:21.20		
Mean	:18.46	Mean	:356.67	Mean	:12.65	Mean	:22.53		
3rd Qu.:	20.20	3rd Qu.:	396.23	3rd Qu.:	16.95	3rd Qu.:	25.00		
Max.	:22.00	Max.	:396.90	Max.	:37.97	Max.	:50.00		

Hide

```
dim(data)
```

```
[1] 506 14
```

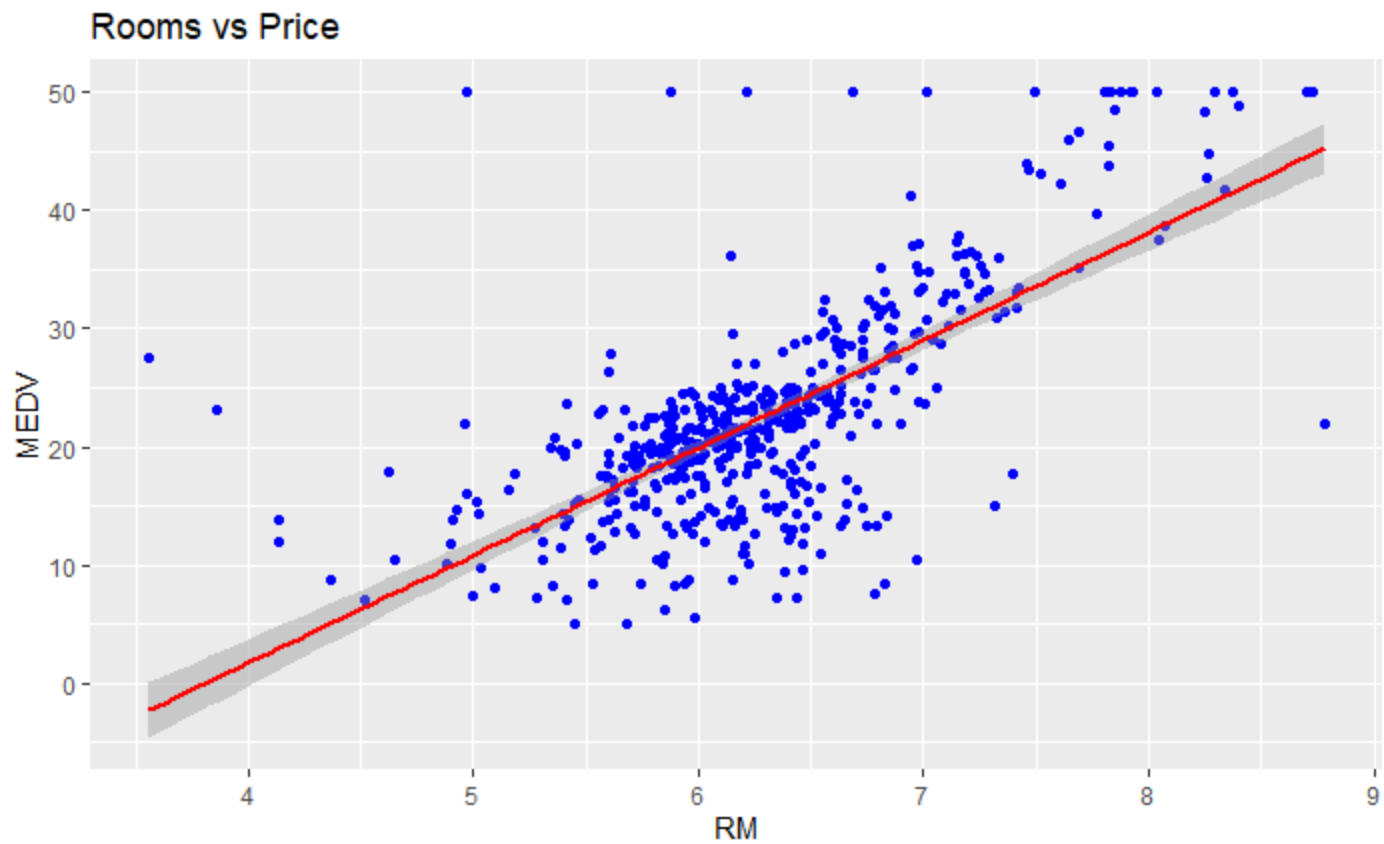
Hide

```
#Missing values column-wise
colSums(is.na(data))
```

CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO
0	0	0	0	0	0	0	0	0	0	0
B	LSTAT	MEDV								
0	0	0								

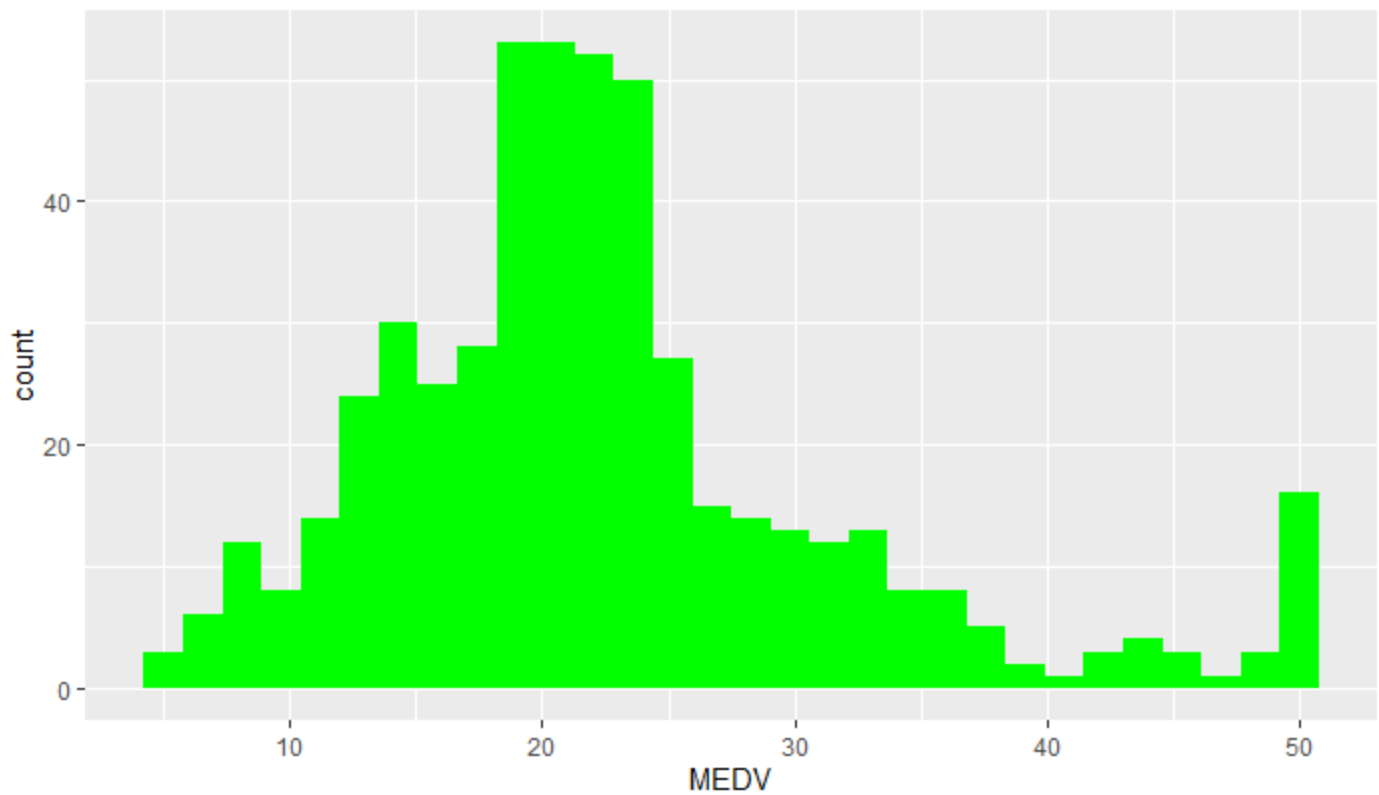
Hide

```
#Scatter plot (Rooms vs Price)
ggplot(data, aes(x = RM, y = MEDV)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red") +
  ggtitle("Rooms vs Price")
```

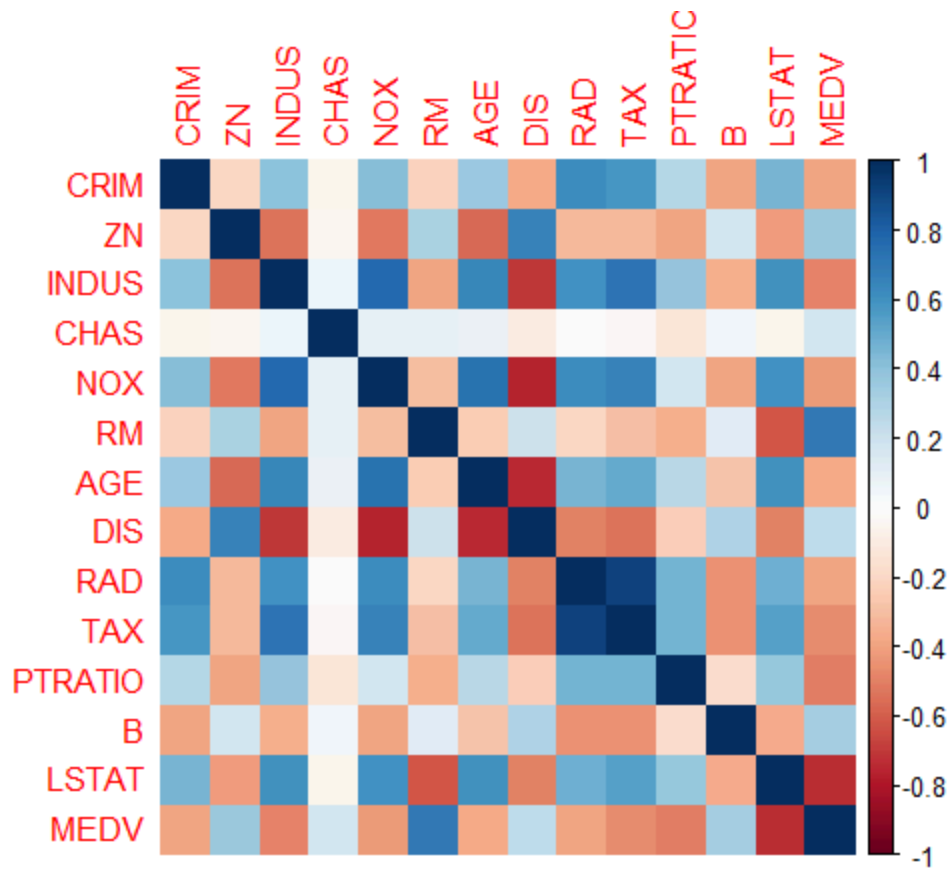
[Hide](#)

```
#Histogram (Price distribution)
ggplot(data, aes(x = MEDV)) +
  geom_histogram(fill = "green", bins = 30) +
  ggtitle("Price Distribution")
```

Price Distribution

[Hide](#)

```
#Correlation heatmap  
cor_matrix <- cor(data)  
corrplot(cor_matrix, method = "color")
```



Hide

```
#Split data (Train/Test)
set.seed(123)

index <- sample(1:nrow(data), 0.8*nrow(data))

train <- data[index, ]
test <- data[-index, ]
```

Hide

```
#Build Model (Linear Regression)
model <- lm(MEDV ~ ., data = train)
summary(model)
```

Call:

```
lm(formula = MEDV ~ ., data = train)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-10.4773	-2.7210	-0.5807	1.7300	24.7911

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	40.136203	5.717257	7.020	9.89e-12	***
CRIM	-0.103364	0.034426	-3.003	0.002850	**
ZN	0.048977	0.015446	3.171	0.001640	**
INDUS	0.018130	0.072197	0.251	0.801856	
CHAS	3.103563	0.923009	3.362	0.000849	***
NOX	-18.599930	4.283486	-4.342	1.80e-05	***
RM	3.432782	0.471435	7.282	1.84e-12	***
AGE	0.011374	0.014906	0.763	0.445897	
DIS	-1.471879	0.222852	-6.605	1.31e-10	***
RAD	0.314074	0.073161	4.293	2.23e-05	***
TAX	-0.011651	0.004267	-2.731	0.006610	**
PTRATIO	-0.986587	0.145275	-6.791	4.16e-11	***
B	0.007513	0.003174	2.367	0.018433	*
LSTAT	-0.588967	0.055680	-10.578	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.731 on 390 degrees of freedom

Multiple R-squared: 0.7441, Adjusted R-squared: 0.7355

F-statistic: 87.21 on 13 and 390 DF, p-value: < 2.2e-16

Hide

#Predictions

```
predictions <- predict(model, newdata = test)
```

Hide

#Model Evaluation (RMSE)

```
rmse <- sqrt(mean((test$MEDV - predictions)^2))
```

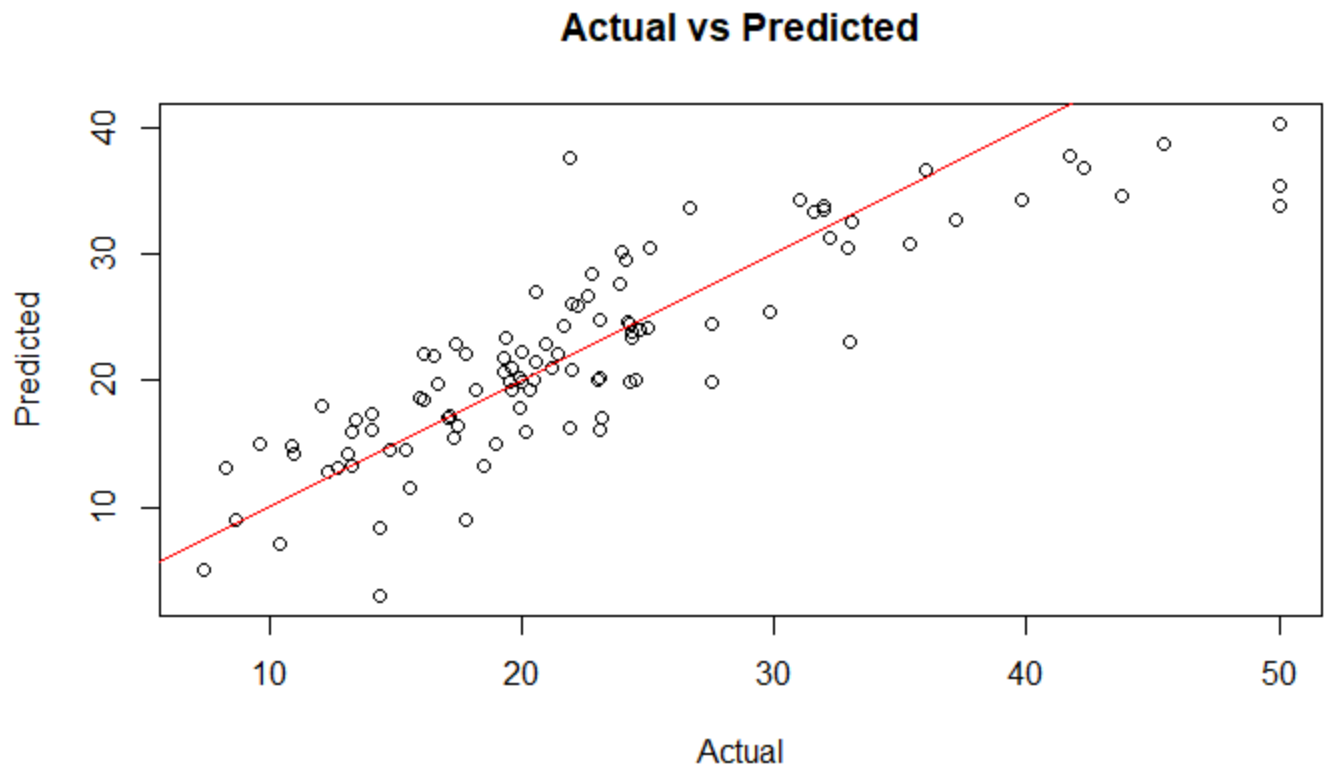
```
rmse
```

```
[1] 4.866069
```

Hide

```
#Actual vs Predicted
plot(test$MEDV, predictions,
     xlab = "Actual",
     ylab = "Predicted",
     main = "Actual vs Predicted")

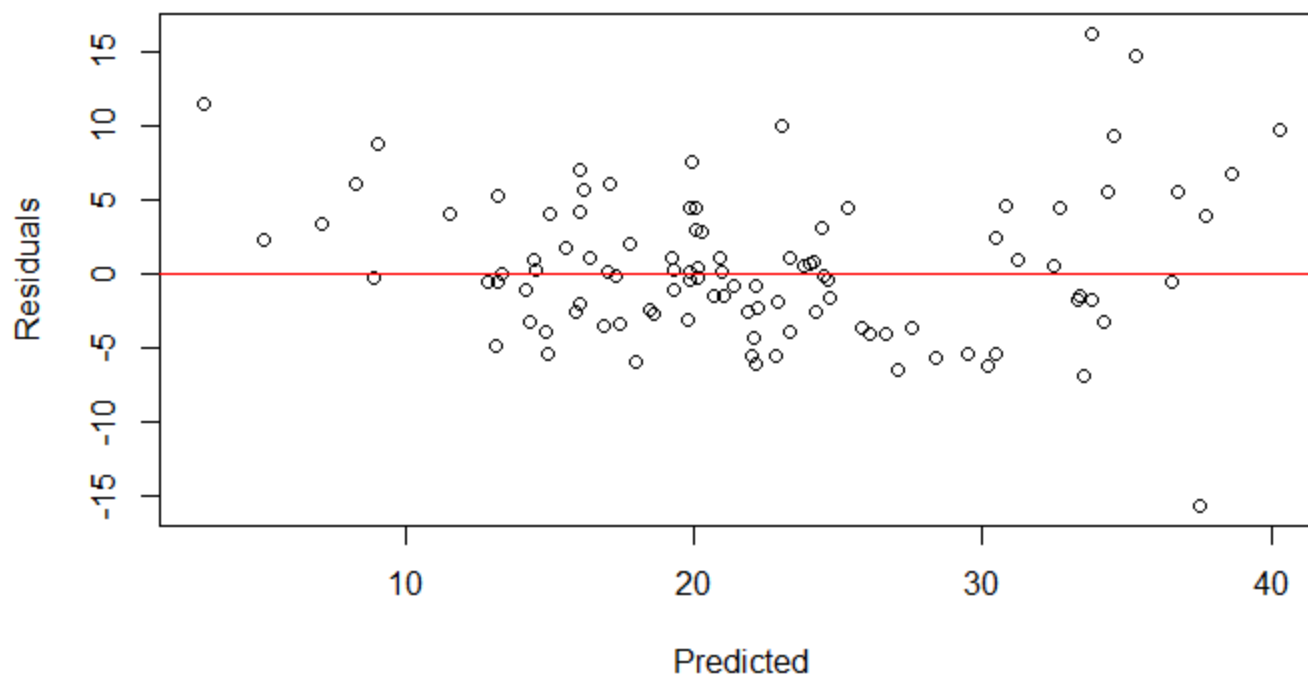
abline(0, 1, col = "red")
```

[Hide](#)

```
#Residual plot
plot(predictions, test$MEDV - predictions,
     xlab = "Predicted",
     ylab = "Residuals",
     main = "Residual Plot")

abline(h = 0, col = "red")
```


Residual Plot

[Hide](#)

#Conclusion (House Price Prediction Model)

cat('This project successfully developed a house price prediction model using multiple linear regression on the Boston housing dataset. The analysis shows that number of rooms (RM) has the strongest positive impact on price, while factors like LSTAT (lower status population), PTRATIO (education quality), and NOX (pollution) negatively affect house values.

The model achieved an R^2 of 0.744, meaning it explains about 74% of the variation in house prices, with an RMSE of ~4.87, indicating reasonably accurate predictions. Visualizations such as scatter plots, heatmaps, and residual plots confirmed meaningful relationships and acceptable model performance.

Overall, the model is effective for predicting house prices and understanding key influencing factors. However, accuracy can be further improved using advanced techniques like Random Forest or XGBoost, especially for handling non-linear relationships.')

This project successfully developed a house price prediction model using multiple linear regression on the Boston housing dataset. The analysis shows that number of rooms (RM) has the strongest positive impact on price, while factors like LSTAT (lower status population), PTRATIO (education quality), and NOX (pollution) negatively affect house values.

The model achieved an R^2 of 0.744, meaning it explains about 74% of the variation in house prices, with an RMSE of ~4.87, indicating reasonably accurate predictions. Visualizations such as scatter plots, heatmaps, and residual plots confirmed meaningful relationships and acceptable model performance.

Overall, the model is effective for predicting house prices and understanding key influencing factors. However, accuracy can be further improved using advanced techniques like Random Forest or XGBoost, especially for handling non-linear relationships.